# 12 Fast Fourier Transform

## Objective

After completing this chapter user will be able to understand

1. Decimation in time algorithm
2. Decimation in frequency algorithm

## Introduction

1. DFT plays an important role in digital signal processing with the help of an efficient algorithm.
2. Fast Fourier Transform (FFT) is simply an efficient method for computing the DFT.
3. The FFT can be used in place of the continuous Fourier Transform only to the extent that the DFT could before, but with a substantial reduction in computer time.
4. Since, most of the problems associated with the use of FFT actually stem from an incomplete or incorrect understanding of the DFT/s.
5. FFT algorithm is used to calculate the DFT when the size N is the power of 2 and when it is the power of 4.
6. FFT is based on the divide and conquers approach, i.e., it uses the decomposition of an N-point DFT into successively smaller DFTs.
7. These algorithms are known as FFT algorithms.

**FFT algorithms:**

1. In this section, we present several methods for computing the DFT efficiency. Basically, the computational problem for the DFT is to compute the sequence X(k) of N complex-valued numbers given another sequence of data x[n] of length N,

   According to the DFT formula,

   $$X(k)=\sum_{n=0}^{N-1}x(n)W_N^{kn} \qquad 0\le k\le N-1$$

   Where $W_N=e^{-j\frac{2\pi}{N}}$

2. In general, the data sequence x[n] is also assumed be complex valued.

   Similarly, the IDFT becomes

   $$x(n)=\frac{1}{N}\sum_{k=0}^{n-1}X(k)W_N^{-kn} \qquad 0\le n\le N-1$$

3. We observe that for each value of k, the direct computation of X(k) involves N complex multiplications (4N real multiplications) and N-1 complex additions (4N – 2 real additions).

   Consequently, to compute all N values of the DFT requires $N^2$ complex multiplications and $N^2-N$ complex additions.

   Symmetry property: $W_N^{k+\frac{N}{2}}=-W_N^k$

   Periodicity property: $W_N^{k+N}=W_N^k$

**Direct computation of the DFT:**

For a complex-valued sequence x[n] of N points, the DFT may be expressed as

$$X_R(k)=\sum_{n=0}^{N-1}\left[X_R[n]\cos\frac{2\pi kn}{N}\times x_I[n]\sin\frac{2\pi kn}{N}\right]$$

$$X_1(k)=\sum_{n=0}^{N-1}\left[X_R[n]\sin\frac{2\pi kn}{N}-x_I[n]\cos\frac{2\pi kn}{N}\right]$$

The direct computation of the above equations requires:

1. $2N^2$ evaluations of trigonometric functions.
2. $4N^2$ real multiplications.
3. 4N (N - 1) real additions.
4. A number of indexing and addressing operations.

**Radix-2 FFT algorithm and decimation in time algorithm:**

1. This approach is very efficient when N is highly composite; that is when N can be factored as $N=r_1r_2r_3...r_v$, where the $\{r_j\}$ are prime. Of particular importance as the case in which $r_1=r_2=...=r_v=r$ so that $N=r^v$

2. In such a case, the DFTs are of size r, so that the computation of the N-point DFT has a regular pattern. The number'r'is called the radix of the FFT algorithm.

Let us consider the computations of an N-point DFT, where N can be factored as a product of two integers, that is, N = LM.

1. Now the sequence x[n], $0 \le n \le N-1$, can be stored in either a one-dimensional array indexed by n or as a two-dimensional array indexed by I and m, where $0 \le I \le L-1$ and $0 \le m \le M-1$.
2. Note that I is the row index, and m is the common index.
3. Thus, the sequence x[n] can be stored in a rectangular array in a variety of ways, each of which depends on the mapping of index n to the indexes (I, m).

n..> 0 1 N-1

| $x[0]$ | $x(1)$ | ... | $x[N-1]$ |
|---|---|---|---|

| row Index \ m | Column Index 0 | 1 | --- | M-1 |
|---|---|---|---|---|
| 0 | x(0,0) | x(0,1) | --- | |
| 1 | x(1,0) | x(1,1) | --- | |
| 2 | x(2,0) | x(2,1) | --- | |
| ⋮ | ⋮ | ⋮ | --- | ⋮ |
| L-1 | | | --- | |

**Fig. 12.1**

For example, suppose that we select the mapping n = MI + m

1. This leads to an arrangement in which the first row consists of the first M elements of x[n], the second row consists of the next M elements of x[n], and so on,
2. On the other hand, the mapping n = I + mL
3. Stores the first L elements of x[n] in the first column, the next L elements in the second column, and so on,

   A similar arrangement can be used to store the computed DFT values. In particular, the mapping is from the index k to a pair of indices (p, q) where $0 \le p \le L-1$ *and* $0 \le q \le M-1$. If we select the mapping k = Mp + q
4. Now suppose that x[n] is mapped into the rectangular array x (l, m) and X(k) is mapped into a corresponding rectangular array X (p, q).
5. Then the DFT can be expressed as a double sum over the elements of the rectangular array multiplied by the corresponding phase factors.

To be specific, let us adopt a column-wise mapping for x[n] and a row-wise mapping for the DFT. Then

$$X(p,q) = \sum_{M=0}^{M-1} \sum_{I=0}^{L-1} X(I,m) W_N^{(Mp+q)(mL+I)}$$

But

$$W_N^{(Mp+q)(mL+I)} = W_n^{MLmp} W_N^{mLq} W_N^{MpI} W_N^{Iq}$$

$$W_N^{Nmp} = 1, W_N^{mpL} = W_{N/L}^{mp} = W_M^{mp},$$

$$And \quad W_N^{MpI} = W_{N/M}^{qL} = W_L^{pI}$$

Let us consider the computation of N= $2^v$

This selection results in a split of the N-point data sequence into two N/2-point data sequences $f_1[n]$ *and* $f_2[n]$, corresponding to the even-numbered and odd-numbered samples of x[n], respectively, that is,

$$f_1[n] = x[2n]$$

$$f_2[n] = x[2n+1], n = 0, 1, \ldots \frac{N}{2} - 1$$

Thus, $f_1[n]$ *and* $f_2[n]$ are obtained by decimating x[n] by a factor of 2, and hence the resulting FFT algorithm is called a decimation-in-time algorithm.

Now the N-point DFT can be expressed in terms of the DFTs of the decimated sequences as follows:

$$X(k) = \sum_{n=0}^{N-1} x[n] W_N^{kn} \quad k = 0, 1, \ldots, N-1$$

$$X(k) = \sum_{n\ even} x[n] W_N^{kn} + \sum_{n\ odd} x[n] W_N^{kn}$$

$$X(k) = \sum_{m=0}^{(N/2)-1} x(2m) W_N^{2mk}$$

$$+ \sum_{m=0}^{(N/2)-1} x(2m+1) W_N^{k(2m+1)}$$

But $W_N^2 = W_{N/2}$ with this substitution,

$$x(k) = \sum_{m=0}^{(N/2)-1} f_1(m) W_{N/2}^{km} + W_N^k \sum_{m=0}^{(N/2)-1} f_2(m) W_{N/2}^{km}$$

$$= F_1(k) + W_N^k F_2(k) \; k = 0, 1, \ldots, N-1$$

Where $F_1(k)$ *and* $F_2(k)$ are the N/2 point DFTs of the sequences $f_1(m)$ *and* $f_2(m)$, respectively.

Since, $F_1(k)$ *and* $F_2(k)$ are periodic, with period N/2, we have $F_1(k + N/2) = F_1(k)$ and $F_2\left(k + \frac{N}{2}\right) = F_2(k)$. In addition, the factor $W_N^{k+N/2} = -W_N^k$.

$$X(k) = F_1(k) + W_N^k F_2(k) \qquad k = 0, 1, \ldots, \frac{N}{2} - 1$$

$$X\left(k + \frac{N}{2}\right) = F_1(k) - W_N^k F_2(k)$$

$$k = 0, 1, \ldots, \frac{N}{2} - 1$$

1. We observed that direct computation of $F_1(k)$ requires $(N/2)^2$ complex multiplications. The same applies to the computation of $F_2(k)$.
2. Furthermore, there is N/2 additional complex multiplication required to compute $W_N^k F_2(k)$.
3. Hence, the computation of X(k) requires $2(N/2)^2 + N/2 = N^2/2 + N/2$ complex multiplications.
4. This first step results in a reduction of the number of multiplications from $N^2$ *to* $N^2/2 + N/2$, which is about a factor of 2 for N large.

   Having performed the decimation-in-time once, we can repeat the process for each of the sequences $f_1[n]$ *and* $f_2[n]$.

**Important points:**

1. The decimation of the data sequence can be repeated again and again until the resulting sequences are reduced to one-point sequences.
2. For $N = 2^v$, this decimation can be performed $v = \log_2 N$ times.
3. Thus, the total number of complex multiplications is reduced to $(N/2)\log_2 N$.
4. The number of complex additions is $N \log_2 N$.

**For radix N butterfly: For calculation of N point DFT:**

1. No. of stages = $\log_2^N$
2. No. of butterfly in each stage = $\frac{N}{2}$
3. Total no. of butterflies = $\frac{N}{2}\log_2 N$
4. Total no. of complex multiplication = $\frac{N}{2}\log_2 N$
5. Total no. of complex addition = $N \log_2 N$

**12.1.** For an N = 8-point DFT. We observe that the computation is performed in three stages, beginning with the computations of four two-point DFTs, then two four-point DFTs, and finally, one eight-point DFT.

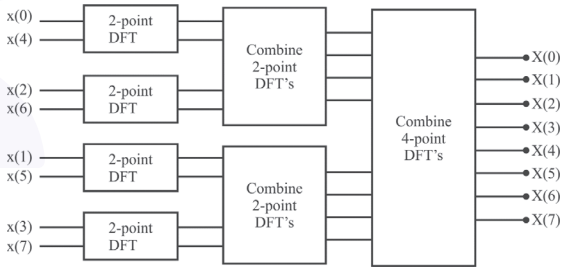



**Fig. 12.2**

The combination of the smaller DFTs to form the larger DFT is illustrated in the below figure for N = 8.

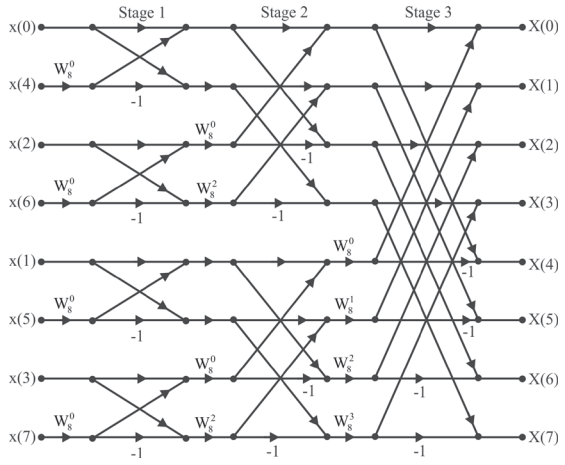



**Fig. 12.3**

**Decimation in frequency algorithm:**

1. Another important radix-2 FFT algorithm, called the decimation in frequency algorithm, is obtained with the choice of M = 2 and L = N/2.
2. This choice of parameters implies column-wise storage of the input data sequence.
3. To drive the algorithm, we begin by splitting the DFT formula into two summations, one of which involves the sum over the first N/2 data points, and the second sum involves the last N/2 data point.
4. The N-point DFT can be expressed in terms of the DFTs of the decimated sequences as follows:

$$X(k)=\sum_{n=0}^{N-1}x[n]W_N^{kn} \qquad k=0,1,\ldots,N-1$$

$$X(k)=\sum_{n=0}^{(N/2)-1}x[n]W_N^{kn}+\sum_{n=N/2}^{N-1}x[n]W_N^{kn}$$

$$=\sum_{n=0}^{(N/2)}x[n]W_N^{kn}+W_N^{kN/2}$$

$$\sum_{n=0}^{(N/2)-1}x\left[n+\frac{N}{2}\right]W_N^{kn}$$

Since, $W_N^{kN/2}=(-1)^k$,

$$X(k)=\sum_{n=0}^{(N/2)-1}\left[x[n]+(-1)^k x\left[n+\frac{N}{2}\right]\right]W_N^{kn}$$

Now, let us split (decimate) X(k) into the even and odd-numbered samples. Thus, we obtain

$$X(2k)=\sum_{n=0}^{(N/2)-1}\left[x(n)+x\left(n+\frac{N}{2}\right)\right]W_{N/2}^{kn}$$

$$k=0,1,\ldots,\frac{N}{2}-1$$

$$x(2k+1)=\sum_{n=0}^{(N/2)-1}\left\{\left[x(n)-x\left(n+\frac{N}{2}\right)\right]W_N^n\right\}W_{N/2}^{kn}$$

$$k=0,1,\ldots,\frac{N}{2}-1$$

Where we have used the fact that $W_N^2=W_{N/2}$

If we define the N/2 point sequence $g_1(n)$ *and* $g_2(n)$ *as*

$$g_1(n)=x(n)+x\left(n+\frac{N}{2}\right)$$

$$g_2(n)=\left[x(n)-x\left(n+\frac{N}{2}\right)\right]W_N^n$$

$$n=0,1,2,\ldots,\frac{N}{2}-1$$

Then $X(2k)=\sum_{n=0}^{(N/2)-1}g_1(n)W_{N/2}^{kn}$

$$X(2k+1)=\sum_{n=0}^{(N/2)-1}g_2(n)W_{N/2}^{kn}$$

a) The computation of the sequences $g_1(n)$ *and* $g_2(n)$ and the subsequent use of these sequences to compute the N/2 - point DFTs are shown in the figure. We observe that the basic computation in this figure involves the butterfly operation.

b) This computational procedure can be repeated through the decimation of the N/2-point DFTs, $X(2k)$ *and* $X(2k+1)$.

c) The entire process involves $v=\log_2 N$

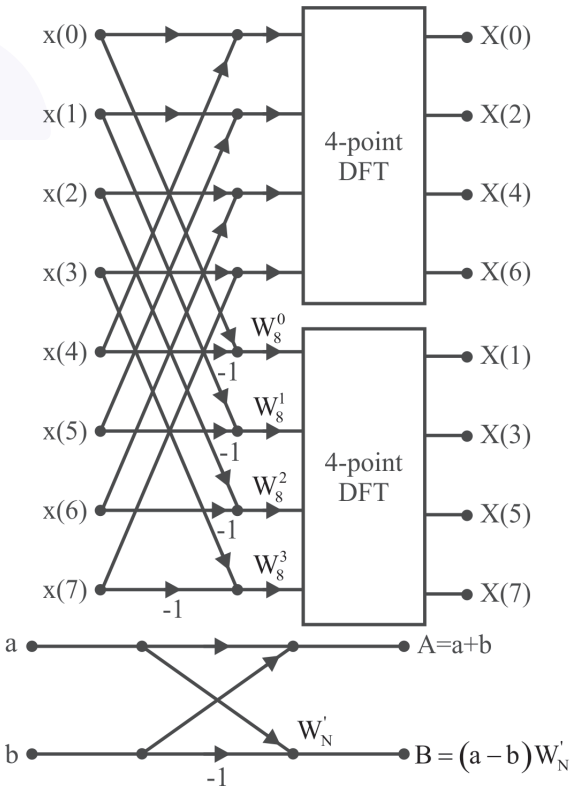



**Fig. 12.4**

d) Stages of decimation, where each stage involves NR butterflies of the type shown in the above figure.

e) Consequently, the computation of the N-point DFT via the decimation in frequency FFT algorithm requires $(N/2)\log_2 N$ complex multiplications and $N\log_2 N$ complex addition, just as in the decimation in time algorithm.

For 8-points, Decimation in Frequency (DIF) algorithm is given by:

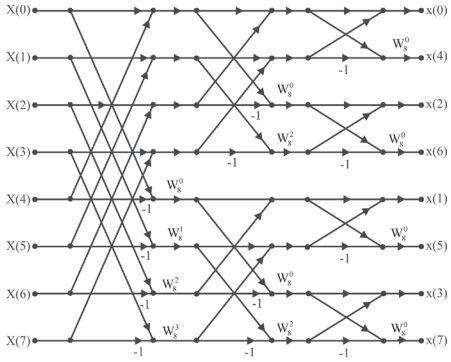



**Fig. 12.5**

Here, the input data x[n] occurs in the natural order, but the output DFT occurs in bit-reversed order.

## Exercise

**1.** For an N-point FFT algorithm with $N = 2^m$, which one of the following statements is true?

(A) It is not possible to construct a signal flow graph with both input and output in normal order.

(B) The number of butterflies in the $m^{th}$ stage is N/m.

(C) In-place computation requires the storage of only 2N node data.

(D) Computation of a butterfly requires only one complex multiplication.

**2.** Consider a six-point Decimation in Time (DIT) Fast Fourier Transform (FFT) algorithm, for which the signal-flow graph corresponding to $X[1]$ is shown in the figure. Let $W_6 = \exp\left(-\frac{j2\pi}{6}\right)$. In the figure, what should be the values of the coefficients $a_1, a_2, a_3$ in terms of $W_6$ so that $X[1]$ is obtained correctly?

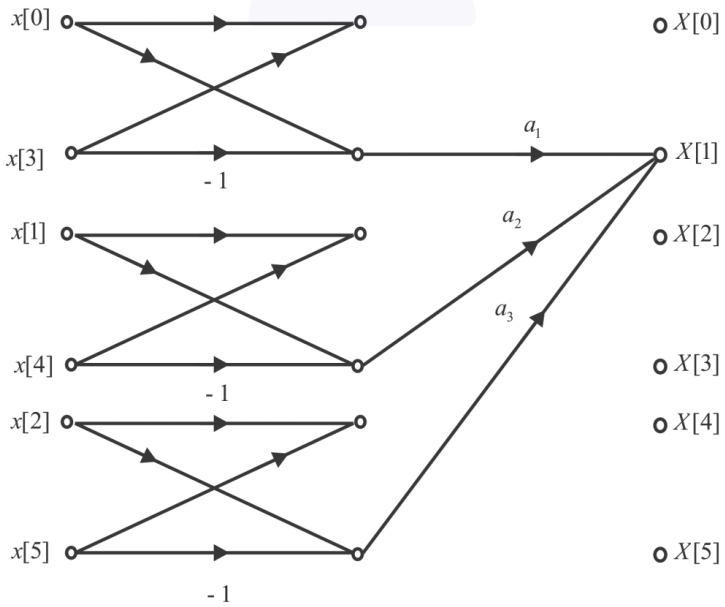



**Fig. 12.6**

(A) $a_1 = -1, a_2 = W_6^2, a_3 = W_6$

(B) $a_1 = 1, a_2 = W_6^2, a_3 = W_6$

(C) $a_1 = 1, a_2 = W_6, a_3 = W_6^2$

(D) $a_1 = -1, a_2 = W_6, a_3 = W_6^2$

**3.** A continuous-time speech signal $x_a(t)$ is sampled at a rate of 8 kHz, and the samples are subsequently grouped in blocks, each of size *N*. The DFT of each block is to be computed in real-time using the radix – 2 Decimation in Frequency FFT algorithm. If the processor performs all operations sequentially, and takes $20\mu s$ for computing each complex multiplication (including multiplications by 1 and -1) and the time required for addition/subtraction is negligible, then the maximum value of *N* is _____.

## Answer Key

| | |
|---|---|
| **1.** | D |
| **2.** | C |
| **3.** | 4096 |

**Table 12.1**

## FFT Solution

**1. Solution: (D)**

Some important points about the N-point FFT algorithm.

1. It is possible to construct a signal flow graph with both input and output in normal order but with additional storage requirements.
2. The number of butterflies required at each stage is $\frac{N}{2}$.
3. In-place computation requires the storage of only N-node data.
4. Computation of a butterfly requires only one complex multiplication and two complex additions.

Hence, the correct option is (D).

**2. Solution: (C)**

To find the 6-point DFT of x[n], Input sequence x[n] is divided into 3 sequences of two samples each.

$$N = p \cdot q$$

$$6 = 3 \times 2$$

$$p = 3, q = 2$$

$$x[n] = \{x(0), x(1), x(2), x(3), x(4), x(5)\}$$

$$f_1(m) = x[3m] = \{x(0), x(3)\} \xleftrightarrow{DFT} F_1(k)$$

$$f_2(m) = x[3m+1] = \{x(1), x(4)\} \xleftrightarrow{DFT} F_2(k)$$

$$f_3(m) = x[3m+2]$$

$$= \{x(2), x(5)\} \xleftrightarrow{DFT} F_3(k)$$

$$X[k] = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$X[k] = \sum_{m=0}^{q-1} x(pm) W_N^{pmk}$$

$$+ \sum_{m=0}^{q-1} x(pm+1) W_N^{(pm+1)k}$$

$$+ \sum_{m=0}^{q-1} x(pm+2) W_N^{(pm+2)k}$$

$$W_N^{pm} = W_6^{3m} = e^{-j\frac{2\pi}{6}\times 3m} = e^{-j\frac{2\pi}{2}\times m} = W_2^m$$

$$\therefore \quad X[k] = \sum_{m=0}^{1} f_1(m) W_2^{km} + W_6 \sum_{m=0}^{1} f_2(m) W_2^{km}$$

$$+ W_6^2 \sum_{m=0}^{1} f_3(m) W_2^{km}$$

$$X[k] = F_1(k) + W_6^k \cdot F_2(k) + W_6^{2k} F_3(k)$$

$$k = 0, 1, 2 \ldots$$

$$F_1(k) = \sum_{m=0}^{1} f_1(m) \cdot W_2^{km}$$

$$F_1(0) = f_1(0) + f_1(1)$$

$$F_1(1) = f_1(0) - f_1(1)$$

$$X(0) = F_1(0) + W_6^0 F_2(0) + W_6^0 F_3(0)$$

$$X(1) = F_1(1) + W_6^1 F_2(1) + W_6^2 F_3(1)$$

$$X(2) = F_1(2) + W_6^2 F_2(2) + W_6^4 F_3(2)$$

$$X(3) = F_1(3) + W_6^3 F_2(3) + W_6^6 F_3(3)$$

$$X(4) = F_1(4) + W_6^4 F_2(4) + W_6^8 F_3(4)$$

$$X(5) = F_1(5) + W_6^5 F_2(5) + W_6^{10} F_3(5)$$

$$W_6^6 = e^{-j\frac{2\pi}{6}\times 6} = 1 = W_6^0$$

$$W_6^8 = W_6^6 \cdot W_6^2 = W_6^2$$

$$[We\ know, W_6^6 = e^{-j\frac{2\pi}{6}\times 6} = e^{-j2\pi} = 1]$$

$$W_6^{10} = W_6^6 \cdot W_6^4 = W_6^4$$

$$Also,\ F_1(k) = F_1(k+2) = F_1(k+4)\ldots$$

$$F_1(0) = F_1(2) = F_1(4)$$

$$F_1(1) = F_1(3) = F_1(5)$$

So, x(0)... are given as

$$X(0) = F_1(0) + F_2(0) + F_3(0)$$

$$X(1) = F_1(1) + W_6^1 F_2(1) + W_6^2 F_3(1)$$

$$X(2) = F_1(0) + W_6^2 F_2(0) + W_6^4 F_3(0)$$

$$X(3) = F_1(1) + W_6^3 F_2(1) + W_6^0 F_3(1)$$

$$X(4) = F_1(0) + W_6^4 F_2(0) + W_6^2 F_3(0)$$

$$X(5) = F_1(1) + W_6^5 F_2(1) + W_6^4 F_3(1)$$

Hence, butterfly structure for N=6 is given as

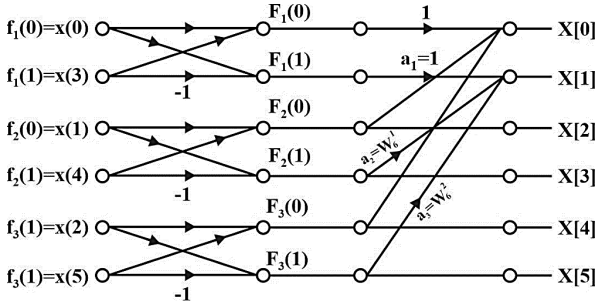



**Fig. 12.7**

Hence, $x(1) = F_1(1) + W_6^1 F_2(1) + W_6^2 F_3(1)$

So, $a_1 = 1, a_2 = W_6^1 \text{ and } a_3 = W_6^2$

Hence, the correct option is (C).

**3. Solution: (4096)**

Given: $f_s = 8000\, samples\ /\ \text{sec}$

$$T_s = \frac{1}{f_s} = \frac{1}{8000}\text{sec}$$

Time for each multiplication $= T_m = 20\mu\text{sec}$ Suppose the block is of size N, then the time is taken to generate each block $= N \times T_{,} = N \times \frac{1}{8000}$

No. of multiplications that can be performed by the processor in the time taken for each block $= \frac{N \times T_s}{T_m}$.

No. of multiplications required to compute DFT by Radix – 2 FFT

$$\text{Algorithm} = \frac{N}{2}\log_2 N$$

No. of multiplication required by FFT ≤ No. of multiplications that can be performed by the processor in the time taken in each block.

$$\frac{N}{2}\log_2 N \le \frac{N \times T_s}{T_m}$$

$$\log_2 N \le \frac{2 \times \frac{1}{8000}}{20 \times 10^{-6}} = 12.5$$

$$N \le 2^{12.5} \le 2^{12} = 4096$$

Hence, the maximum value of N is 4096.